

TESTS FOR THE FUNCTIONAL GROUPS IN

ORGANIC COMPOUND - 1:

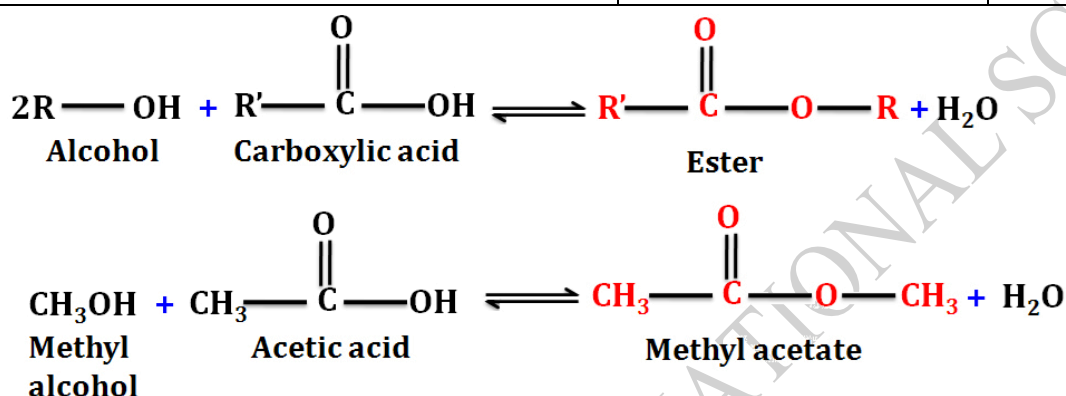
Experiment	Observation	Inference
Tests for Unsaturation: Dissolve a little of the given organic compound in bromine water to it and shake well.	No disappearance of orange-red colour.	Compound is saturated.
Baeyer's Test (Alkaline KMnO₄ Test): Dissolve a little of the given organic compound in few drops of alkaline solution of KMnO ₄ to it and shake well.	No disappearance of pink colour of KMnO ₄ .	Compound is saturated.
Tests for Alcoholic Group:		
1. Ceric ammonium nitrate Test: Take about 1 ml of the given liquid and add few drops of ceric ammonium nitrate reagent and shake it well.	Pink or red colour is developed	Presence of alcoholic group.
$ \begin{array}{l} 2R - OH + (NH_4)_2Ce(NO_3)_6 \longrightarrow (ROH)_2Ce(NO_3)_4 + 2NH_4NO_3 \\ \text{Alcohol} \qquad \text{Ceric ammonium nitrate} \qquad \text{Alkoxy cerium (IV) compound (Pink or Red colour)} \\ \\ 2CH_3 - OH + (NH_4)_2Ce(NO_3)_6 \longrightarrow (CH_3OH)_2Ce(NO_3)_4 + 2NH_4NO_3 \\ \text{Methyl alcohol} \qquad \text{Ceric ammonium nitrate} \qquad \text{Methoxy cerium (IV) compound (Pink or Red colour)} \end{array} $		

2.Ester Test:

Take about 1 ml of the give liquid, add about 1 ml glacial acetic acid and 2-3 drops of conc. sulphuric acid. Warm the mixture on a water bath for about 10 minutes. Pour it into about 20 ml of cold water taken in a beaker and smell.

A fruity smell is produced.

Presence of alcoholic group.

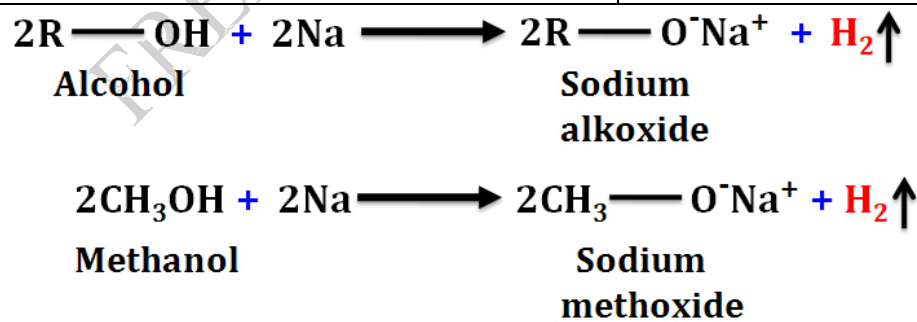


3.Sodium metal Test:

Take about 1 ml of the given liquid, add about 1gram of anhydrous calcium sulphate and shake well to remove water. Filter the liquid into another clean dry test tube and add a small piece of sodium metal.

Brisk effervescence is produced.

Brisk effervescence is due to the evolution of hydrogen gas indicating the presence of alcoholic group.

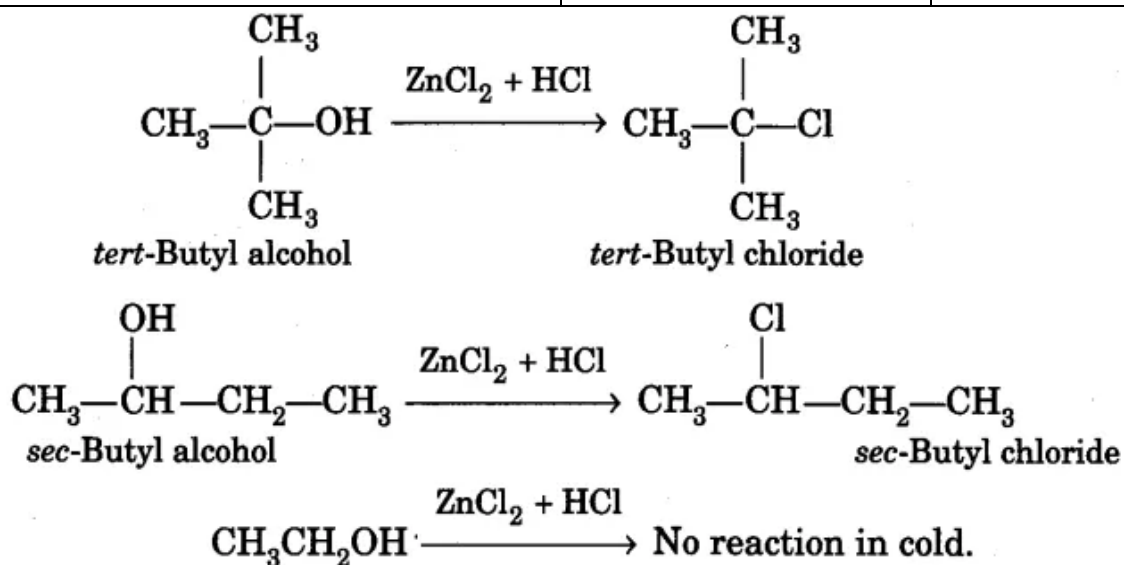


4. Lucas Test:

Take about 1 ml of dry alcohol in a clean dry test tube and add 8-10 drops of Lucas reagent. Shake the mixture well.

Cloudiness appears only upon heating.

Presence of primary alcohol.



The given organic compound contains alcohol as functional group.

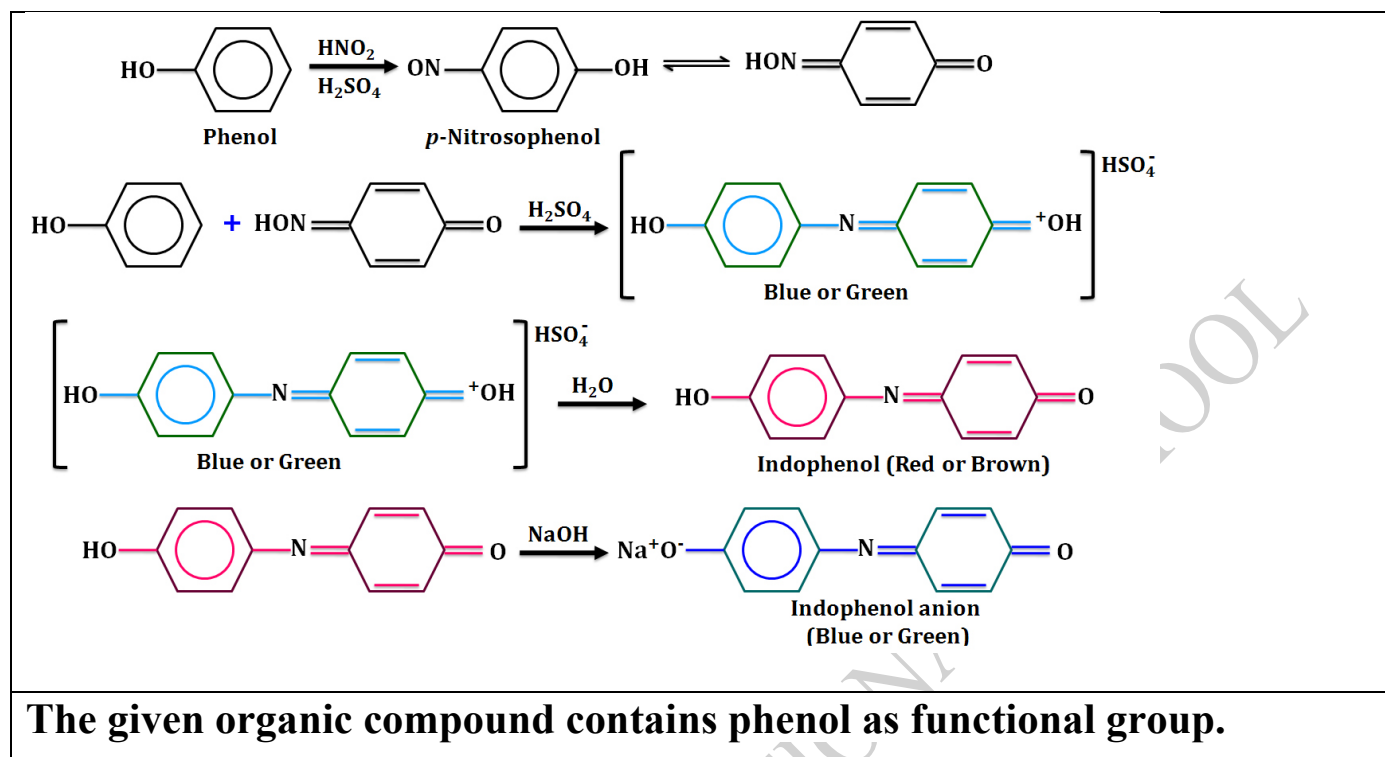
TESTS FOR THE FUNCTIONAL GROUPS IN

ORGANIC COMPOUND - 2:

Experiment	Observation	Inference
Tests for Unsaturation: Dissolve a little of the given organic compound in bromine water to it and shake well.	Disappearance of orange-red colour of bromine.	Compound is unsaturated.
Baeyer's Test (Alkaline KMnO₄ Test): Dissolve a little of the given organic compound in few drops of alkaline solution of KMnO ₄ to it and shake well.	Disappearance of pink colour of KMnO ₄ .	Compound is unsaturated.
Tests for Alcoholic Group:		
Ceric ammonium nitrate Test: Take about 1 ml of the given liquid and add few drops of ceric ammonium nitrate reagent and shake it well.	No pink or red colour	Alcoholic group is absent.

Tests for Phenolic Group		
1. Ferric chloride Test: Take about 1 ml of neutral ferric chloride solution, add 2-3 drops of the given liquid (or crystal is solid).	Violet colour is seen.	Presence of phenolic group.

$\text{FeCl}_3 + 6\text{C}_6\text{H}_5\text{OH} \longrightarrow [\text{Fe}(\text{OC}_6\text{H}_5)_6]^{3+} + 3\text{HCl}$ Ferric chloride Phenol Violet complex		
2. Litmus Test: Place a drop of the given liquid (or a crystal if solid) on a moist blue litmus paper.	Blue litmus turns to red.	Presence of phenolic group.
Phenol Phenoxide ion Hydronium ion		
3. Liebermann's Test: Take a small quantity of sodium nitrite and add about 1 ml of the given liquid. Heat the test tube very gently for half a minute and allow it to cool. Then add about 1 ml of conc. sulphuric acid and shake the test tube to mix the contents. Add distilled water carefully to the test tube. Add excess of sodium hydroxide solution to the test tube.	A deep blue or green colouration is seen. Colour turns to red. Deep blue or green colour appears.	Presence of phenolic group.



TESTS FOR THE FUNCTIONAL GROUPS IN

ORGANIC COMPOUND - 3:

Experiment	Observation	Inference
Tests for Unsaturation: Dissolve a little of the given organic compound in bromine water to it and shake well.	No disappearance of orange-red colour.	Compound is saturated.
Baeyer's Test (Alkaline KMnO₄ Test): Dissolve a little of the given organic compound in few drops of alkaline solution of KMnO ₄ to it and shake well.	No disappearance of pink colour of KMnO ₄ .	Compound is saturated.
Tests for Alcoholic Group:		
Ceric ammonium nitrate Test: Take about 1 ml of the given liquid and add few drops of ceric ammonium nitrate reagent and shake it well.	No pink or red colour	Alcoholic group is absent.
Tests for Phenolic Group		
Ferric chloride Test: Take about 1 ml of neutral ferric chloride solution, add 2-3 drops of the given liquid (or crystal is solid).	No violet colour is seen.	Phenolic group is absent.
Tests for Aldehydic and Ketonic Groups		

1. 2,4-Dinitrophenyl hydrazine Test: Take a small quantity of the given compound, add few drops of 2, 4-dinitrophenyl hydrazine solution, shake the mixture and allow it to stand for 5 minutes.	Yellow or orange precipitate is formed.	Presence of carbonyl group.
$\text{CH}_3 - \overset{\text{H}}{\underset{\text{O}}{\text{C}}} = \text{O} + \text{H}_2\text{N} - \text{HN} - \text{C}_6\text{H}_3(\text{NO}_2)_2 \longrightarrow \text{CH}_3 - \overset{\text{H}}{\underset{\text{N}}{\text{C}}} = \text{N} - \text{HN} - \text{C}_6\text{H}_3(\text{NO}_2)_2$ Acetaldehyde 2,4-Dinitrophenyl hydrazine Acetaldehyde-2,4-dinitrophenyl hydrazone $\text{CH}_3 - \overset{\text{CH}_3}{\underset{\text{O}}{\text{C}}} = \text{O} + \text{H}_2\text{N} - \text{HN} - \text{C}_6\text{H}_3(\text{NO}_2)_2 \longrightarrow \text{CH}_3 - \overset{\text{CH}_3}{\underset{\text{N}}{\text{C}}} = \text{N} - \text{HN} - \text{C}_6\text{H}_3(\text{NO}_2)_2$ Acetone 2,4-Dinitrophenyl hydrazine Acetone-2,4-dinitrophenyl hydrazone		
2. Sodium bisulphite Test: Take about 2 ml of saturated solution of sodium bisulphite, add a small quantity of the given compound. Shake and leave it for 15-20 minutes.	White crystalline precipitate is formed.	Presence of carbonyl group.
$\text{R} - \overset{\text{H}}{\underset{\text{O}}{\text{C}}} = \text{O} + \text{NaHSO}_3 \longrightarrow \text{R} - \overset{\text{H}}{\underset{\text{SO}_3^-\text{Na}^+}{\text{C}}} - \text{OH}$ Aldehyde Sodium bisulphite Bisulphite addition product $\text{R} - \overset{\text{R}}{\underset{\text{O}}{\text{C}}} = \text{O} + \text{NaHSO}_3 \longrightarrow \text{R} - \overset{\text{H}}{\underset{\text{SO}_3^-\text{Na}^+}{\text{C}}} - \text{OH}$ Aldehyde Sodium bisulphite Bisulphite addition product		

Differentiating Tests for Aldehydes 1. Tollen's Test (Silver mirror test): 3-4 drops of the given liquid (or 0.1 g if solid) + Tollen's reagent and warm the test tube on a water bath for about 5 minutes.	Shining silver mirror is formed.	Presence of aldehydic group is confirmed.
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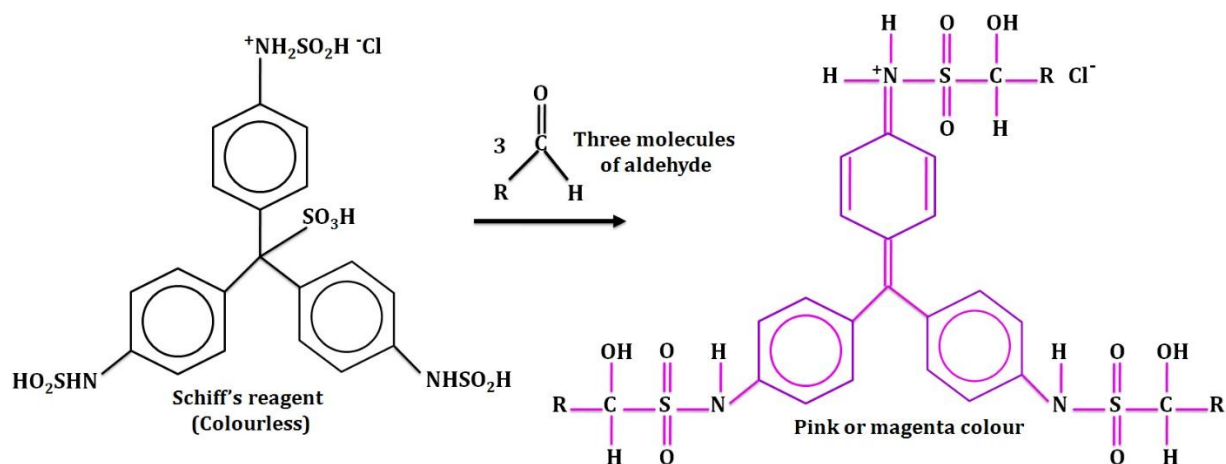
$ \begin{array}{c} \text{O} \\ \\ \text{R} - \text{C} - \text{H} \\ \text{Aldehyde} \end{array} + 2\text{Ag}(\text{NH}_3)_2\text{OH} \longrightarrow 2\text{Ag(s)} + \begin{array}{c} \text{O} \\ \\ \text{R} - \text{C} - \text{O}^- \text{NH}_4^+ \end{array} + \text{H}_2\text{O} + 3\text{NH}_3 $ Tollen's reagent Silver		
2. Fehling's Test: Take 1 ml each of Fehling's solution A and B in a test tube. Add 4-5 drops of the given organic liquid (or 0.2g if solid) and warm the test in a water bath for 4-5 minutes.	Red precipitate is formed.	Presence of aldehydic group is confirmed.
$ \text{RCHO} + \underbrace{2\text{Cu}^{2+} + 5\text{OH}^-}_{\text{Fehling's solution}} \longrightarrow \text{Cu}_2\text{O (s)} + \text{RCOO}^-(\text{s}) + 3\text{H}_2\text{O} $ Aldehyde Cuprous oxide (Red)		

3. Schiff's Test:

Dissolve a small quantity of the given compound in alcohol, add about 1 ml of Schiff's reagent and shake it well.

Pink colour is seen.

Presence of aldehydic group is confirmed.



The given organic compound contains aldehyde as functional group.

TESTS FOR THE FUNCTIONAL GROUPS IN

ORGANIC COMPOUND - 4:

Experiment	Observation	Inference
Tests for Unsaturation: Dissolve a little of the given organic compound in bromine water to it and shake well.	No disappearance of orange-red colour.	Compound is saturated.
Baeyer's Test (Alkaline KMnO_4 Test): Dissolve a little of the given organic compound in few drops of alkaline solution of KMnO_4 to it and shake well.	No disappearance of pink colour of KMnO_4 .	Compound is saturated.
Tests for Alcoholic Group:		
Ceric ammonium nitrate Test: Take about 1 ml of the given liquid and add few drops of ceric ammonium nitrate reagent and shake it well.	No pink or red colour	Alcoholic group is absent.
Tests for Phenolic Group		
Ferric chloride Test: Take about 1 ml of neutral ferric chloride solution, add 2-3 drops of the given liquid (or crystal is solid).	No violet colour is seen.	Phenolic group is absent.
Tests for Aldehydic and Ketonic Groups		

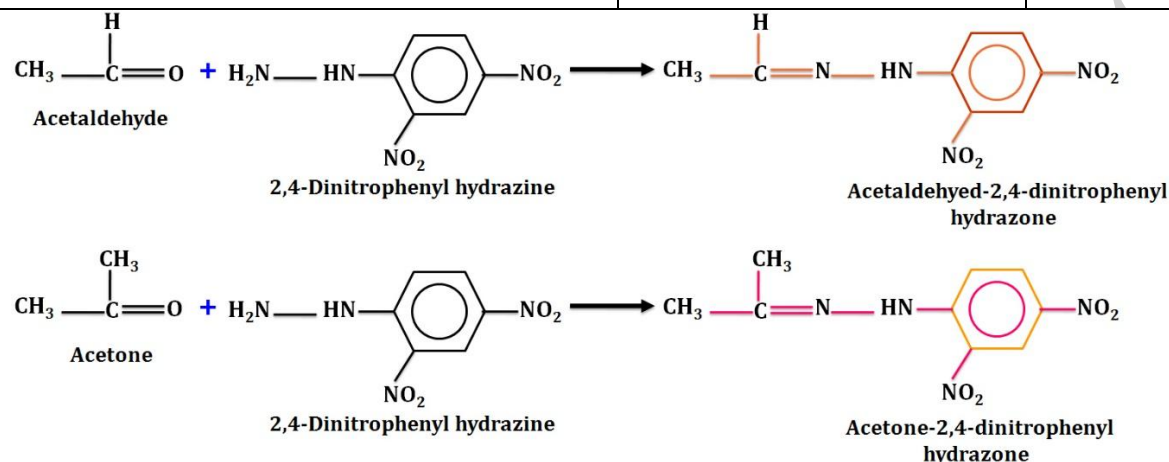
2,4-Dinitrophenyl hydrazine

Test:

Take a small quantity of the given compound, add few drops of 2, 4-dinitrophenyl hydrazine solution, shake the mixture and allow it to stand for 5 minutes.

Yellow or orange precipitate is formed.

Presence of carbonyl group.

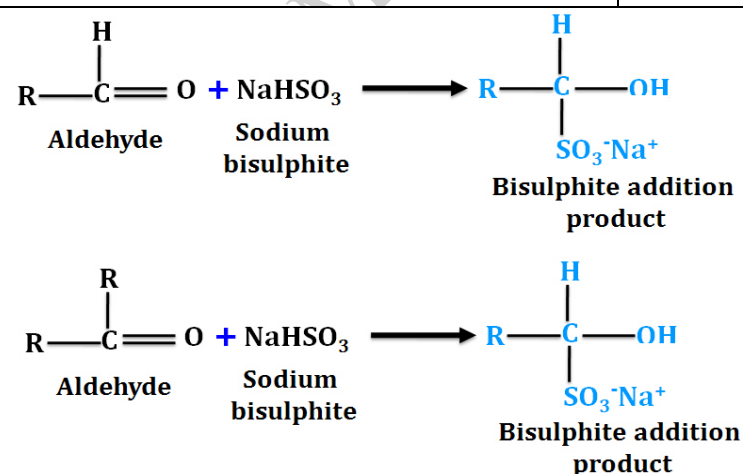


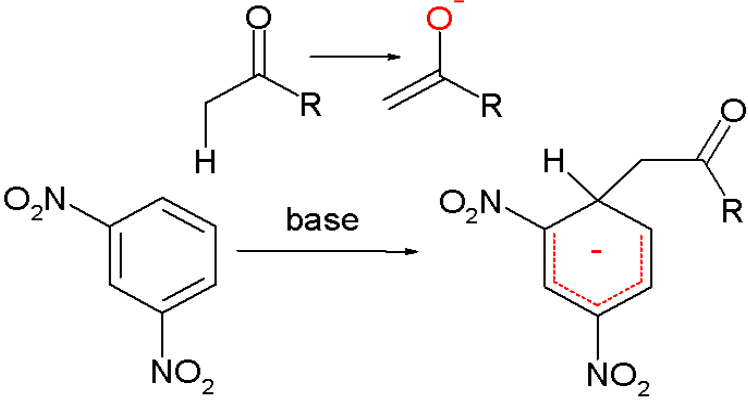
Sodium bisulphite Test:

Take about 2 ml of saturated solution of sodium bisulphite, add a small quantity of the given compound. Shake and leave it for 15-20 minutes.

White crystalline precipitate is formed.

Presence of carbonyl group.



Differentiating Tests for Aldehydes Tollen's Test (Silver mirror test): 3-4 drops of the given liquid (or 0.1 g if solid) + Tollen's reagent and warm the test tube on a water bath for about 5 minutes.	No Shining silver mirror formed.	Aldehydic group is absent.
Differentiating Tests for Ketones 1. <i>m</i>-Dinitrobenzene Test: Take a small quantity of the given compound, add about 0.1g of finely powdered <i>m</i>-dinitrobenzene. Now add about 1 ml of dilute sodium hydroxide solution and shake it well.	Violet colour is formed which slowly fades away.	Presence of ketonic group is confirmed.
		

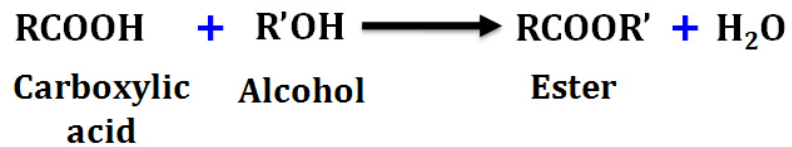
2. Sodium nitroprusside Test: Dissolve a small quantity of sodium nitroprusside in about 1 ml of distilled water, add a small quantity of the given compound. Shake the test tube well and add sodium hydroxide solution dropwise.	Red colouration.	Presence of ketonic group is confirmed.
$\text{CH}_3\text{COCH}_3 + \text{OH}^- \longrightarrow \text{CH}_3\text{COCH}_2^- + \text{H}_2\text{O}$ $[\text{Fe}(\text{CN})_5\text{NO}]^{2-} + \text{CH}_3\text{COCH}_3 \longrightarrow [\text{Fe}(\text{CN})_5\text{NO} \cdot \text{CH}_3\text{COCH}_2]^{3-}$ Nitroprusside ion Red colouration		
The given organic compound contains ketone as functional group.		

TESTS FOR THE FUNCTIONAL GROUPS IN

ORGANIC COMPOUND - 5:

Experiment	Observation	Inference
Tests for Unsaturation: Dissolve a little of the given organic compound in bromine water to it and shake well.	No disappearance of orange-red colour.	Compound is saturated.
Baeyer's Test (Alkaline KMnO_4 Test): Dissolve a little of the given organic compound in few drops of alkaline solution of KMnO_4 to it and shake well.	No disappearance of pink colour of KMnO_4 .	Compound is saturated.
Tests for Alcoholic Group:		
Ceric ammonium nitrate Test: Take about 1 ml of the given liquid and add few drops of ceric ammonium nitrate reagent and shake it well.	No pink or red colour	Alcoholic group is absent.
Tests for Phenolic Group		
Ferric chloride Test: Take about 1 ml of neutral ferric chloride solution, add 2-3 drops of the given liquid (or crystal is solid).	No violet colour is seen.	Phenolic group is absent.
Tests for Aldehydic and Ketonic Groups		

2,4-Dinitrophenyl hydrazine Test: Take a small quantity of the given compound, add few drops of 2, 4-dinitrophenyl hydrazine solution, shake the mixture and allow it to stand for 5 minutes.	No yellow or orange precipitate.	Carbonyl group is absent.
Tests for Carboxylic Group		
1. Litmus Test: Place a drop of the given liquid (or a crystal if solid) on a moist blue litmus paper. 2. Sodium Hydrogen Carbonate Test: Take 1 ml of the given organic compound, add a pinch of sodium bicarbonate to it.	Blue litmus turns to red. Brisk effervescence is observed.	Presence of carboxylic group. Presence of carboxylic group.
$\text{RCOOH} + \text{NaHCO}_3 \longrightarrow \text{RCOONa} + \text{CO}_2 (\text{g}) + \text{H}_2\text{O}$ Carboxylic acid Brisk effervescence		
3. Ester Test: Take a small quantity of the given organic compound, add 1 ml of ethyl alcohol and 1-2 drops of conc. sulphuric acid into it. Heat the reaction mixture on a water bath for about 5 minutes. Pour the mixture into a beaker containing water.	Fruity smell is observed.	Presence of carboxylic group.



The given organic compound contains carboxylic acid as functional group.

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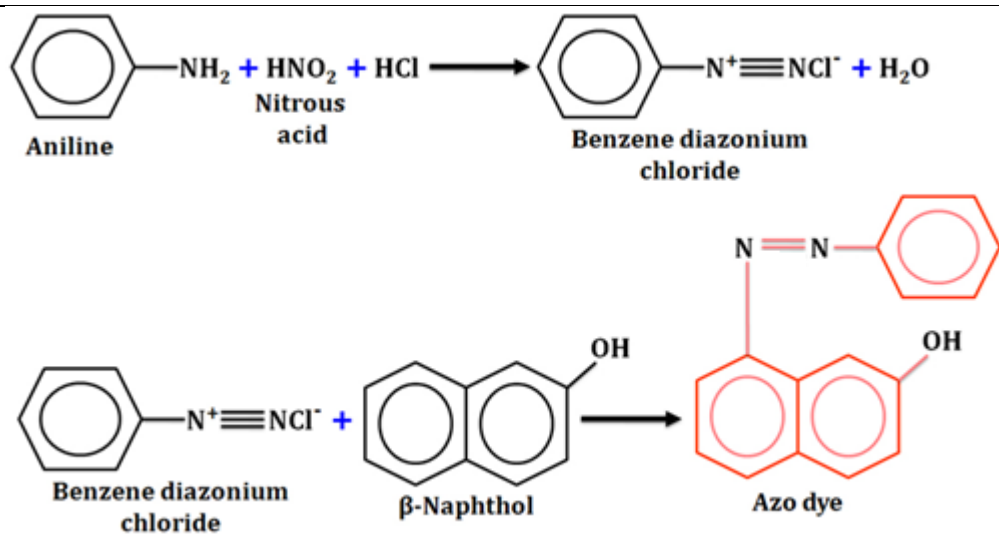
TESTS FOR THE FUNCTIONAL GROUPS IN

ORGANIC COMPOUND - 6:

Experiment	Observation	Inference
Tests for Unsaturation: Dissolve a little of the given organic compound in bromine water to it and shake well.	Disappearance of orange-red colour of bromine.	Compound is unsaturated.
Baeyer's Test (Alkaline KMnO_4 Test): Dissolve a little of the given organic compound in few drops of alkaline solution of KMnO_4 to it and shake well.	Disappearance of pink colour of KMnO_4 .	Compound is unsaturated.
Tests for Alcoholic Group:		
Ceric ammonium nitrate Test: Take about 1 ml of the given liquid and add few drops of ceric ammonium nitrate reagent and shake it well.	No pink or red colour	Alcoholic group is absent.
Tests for Phenolic Group		
Ferric chloride Test: Take about 1 ml of neutral ferric chloride solution, add 2-3 drops of the given liquid (or crystal is solid).	No violet colour is seen.	Phenolic group is absent.
Tests for Aldehydic and Ketonic Groups		

2,4-Dinitrophenyl hydrazine Test: Take a small quantity of the given compound, add few drops of 2, 4-dinitrophenyl hydrazine solution, shake the mixture and allow it to stand for 5 minutes.	No yellow or orange precipitate.	Carbonyl group is absent.
Tests for Carboxylic Group		
Litmus Test: Place a drop of the given liquid (or a crystal if solid) on a moist blue litmus paper.	No change is observed.	Carboxylic group is absent.
Tests for Amino Group		
1. Solubility Test: Take a small quantity of the given compound and add 2-3 ml of dil.HCl and shake the test tube well.	Compound is soluble in dil.HCl.	Compound may be amine.
$\text{C}_6\text{H}_5\text{NH}_2 + \text{HCl} \longrightarrow \text{C}_6\text{H}_5\text{NH}_3^+\text{Cl}^-$ Aniline Anilinium chloride		
2. Litmus Test: Place a drop of the given liquid (or a pinch of solid) on a moist red litmus paper.	Red litmus turns to blue.	Presence of primary amine.

3. Carbylamines Test: Take a small quantity of solid KOH and add about 2 ml of ethanol. Warm the test tube until the pellets dissolve. To this add a few drops of chloroform and small amount of the given compound and warm gently.	An offensive smell is produced.	Presence of primary amine.
$\text{RNH}_2 + \text{CHCl}_3 + 3\text{KOH} \longrightarrow \text{RNC} + 3\text{KCl} + 3\text{H}_2\text{O}$ Primary amine Isocyanide		
4. Azo-Dye Test: Take three test tubes A, B and C. In test tube A, dissolve a small quantity of the compound in 2 ml of HCl. In test tube B, prepare an aqueous solution of sodium nitrite. In test tube C, dissolve a small quantity of β-naphthol in dilute sodium hydroxide. Place all the three test tubes in an ice bath. Now add sodium nitrite solution into test tube A and the resulting solution is added to test tube C.	Formation of a red or orange dye.	Presence of aromatic primary amine is confirmed.



The given organic compound contains amine as functional group.